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The Impact of Elastic Deformation on Deposition Kinetics at Lithium/Polymer Interfaces

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Past theories of electrode stability assume that the surface tension resists the amplification of surface roughness at cathodes and show that instability at lithium/liquid interfaces cannot be prevented by surface forces alone [Electrochim. Acta, **40**, 599 (1995)]. This work treats interfacial stability in lithium/polymer systems where the electrolyte is solid. Linear elasticity theory is employed to compute the additional effect of bulk mechanical forces on electrode stability. The lithium and polymer are treated as Hookean elastic materials, characterized by their shear moduli and Poisson's ratios. Two-dimensional displacement distributions that satisfy force balances across a periodically deforming interface are derived; these allow computation of the stress and surface-tension forces. The incorporation of elastic effects into a kinetic model demonstrates regimes of electrolyte mechanical properties where amplification of surface roughness can be inhibited. For a polymer material with Poisson's ratio similar to poly(ethylene oxide), interfacial roughening is mechanically suppressed when the separator shear modulus is about twice that of lithium.
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Dendrite formation presents a major obstacle to the development of practical lithium-electrode batteries. Active research into polymer electrolytes has persisted for nearly 30 years on the basis that these materials are found to strengthen cells mechanically and improve the cyclability of lithium electrodes.[1-4] Typical continuum-scale models of dendrite growth assume that surface tension resists the natural tendency of electrodes to roughen during deposition and do not incorporate other mechanical effects.[5,6]

Recently a kinetic model that includes the effects of general stresses on current density distributions at roughening interfaces was proposed.[1] The model requires that stress distributions be obtained by solution of the steady-state equation of motion. Stress distributions are computed here by treating the electrode and separator as isotropic linear-elastic (Hookean solid) materials. Such materials are characterized by the Poisson's ratio and shear modulus, which are properties that result from straightforward rheometric experiments.

In this paper, the interface between the electrode and electrolyte is subjected to an idealized two-dimensional displacement of arbitrary amplitude and frequency. A unique solution to the equation of motion is specified with additional boundary conditions required by the proposed kinetic model. In the regime of small-amplitude two-dimensional perturbations to the interface, analytical solutions are obtained for displacement distributions; these results are used to compute deformation profiles on either side of the interface.

Known deformation profiles allow computation of the compressive stress, deformation stress, and surface tension forces that develop at an elastic interface in response to a small-amplitude periodic disturbance. Incorporation of these forces into the kinetic model demonstrates that they contribute to changes in exchange current densities along the roughening surface. Variation of the separator mechanical properties allows prediction of conditions under which the instability of the lithium/polymer interface is eliminated, *i.e.*, circumstances where surface roughening does not amplify and dendrites may be prevented. This regime of stability is shown to occur at shear moduli three or more orders of magnitude higher than those obtained thus far in even the sturdiest polymer electrolytes, which have moduli in the megapascal range.[7-9] Some qualitative justification for the improved resistance to dendrites exhibited by current lithium/polymer systems is also provided by the present theoretical approach.

Theoretical

In a previous communication,[1] the slightly differing kinetic relationships employed by Barton and Bockris,[6] Diggle *et al.*,[10] Oren and Landau,[11] and Sundström and Bark[5] were reconciled and generalized to include mechanical forces due to viscous and deformational stresses and local concentration variation at the electrode. For the sake of simplicity in our exploration of mechanical effects, we neglect local concentration variation and assume the electrolyte is 1:1 (LiX), with an elementary electrode reaction

$$\mathrm{Li} \leftrightarrow \mathrm{Li}^+ + \mathrm{e}^- \qquad [1]$$

With these simplifications incorporated into the general mechanism, the reaction kinetics are found to be described by a modified Butler-Volmer equation of the form[1]

$$i_\mathrm{n} = i_{0,\mathrm{ref}} \exp\left[\frac{(1-\alpha_\mathrm{a})\Delta\mu_{\mathrm{e}^-}}{RT}\right]\left[\exp\left(\frac{\alpha_\mathrm{a}F\eta_\mathrm{s}}{RT}\right) - \exp\left(-\frac{\alpha_\mathrm{c}F\eta_\mathrm{s}}{RT}\right)\right] \qquad [2]$$

where R is the gas constant, F is Faraday's constant, T is the temperature, η_s is the surface overpotential, i_n is current density normal to the deformed interface, $i_{0,\mathrm{ref}}$ is the exchange current density at an undeformed interface, α_a and α_c are anodic and cathodic transfer coefficients, respectively, and $\Delta\mu_{\mathrm{e}^-}$ is the electrochemical potential change in the electrons within the electrode induced by local strain or interfacial shape change. In proposing Eq. 2, we have followed Barton and Bockris in the assumption that the resistance to charge transfer is on the solution side of the interface.

The deformation parameter $\Delta\mu_{\mathrm{e}^-}$ can be expressed in terms of the thermodynamic properties of the electrode and solution and the forces acting on the interface as[1]

$$\Delta\mu_{\mathrm{e}^-} = -\frac{1}{2}(\overline{V}_\mathrm{Li} + t_-^0\overline{V}_\mathrm{LiX}) \times \{-\gamma\overline{\nabla}_\mathrm{s}\cdot\mathbf{e}_\mathrm{n} + \mathbf{e}_\mathrm{n}\cdot[\mathbf{e}_\mathrm{n}\cdot(\underline{\underline{\tau}}_\mathrm{d}^{\mathrm{electrode}} - \underline{\underline{\tau}}_\mathrm{d}^{\mathrm{separator}})]\} + \frac{1}{2}(\overline{V}_\mathrm{Li} - t_-^0\overline{V}_\mathrm{LiX})(\Delta p^{\mathrm{electrode}} + \Delta p^{\mathrm{separator}}) \qquad [3]$$

where $\overline{V}_\mathrm{Li}$ is the molar volume of the electrode, $\overline{V}_\mathrm{LiX}$ is the partial molar volume of the electrolyte salt, t_-^0 is the anion transference number, γ is the surface energy, $\underline{\underline{\tau}}_\mathrm{d}^\mathrm{i}$ and Δp^i are the deformation stress tensor and the gage pressure in phase i, respectively, $\mathbf{e}_\mathrm{n}$ is the unit surface normal vector pointing into the electrolytic solution, and $\overline{\nabla}_\mathrm{s}$ is the surface gradient. The partial molar volumes, transference number, and surface energy are material properties; the deformation stresses and pressures in Eq. 3 can be obtained by solution of the steady-state equation of motion

$$-\overline{\nabla}\cdot\underline{\underline{\sigma}} = 0 \qquad [4]$$

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in each phase. In an equilibrated elastic solid, the total stress $\underline{\underline{\sigma}}$ is related to the deformation stress and gage pressure by

$$\underline{\underline{\sigma}} = \underline{\underline{\tau}}_d + p\underline{\underline{I}} \quad [5]$$

where $\underline{\underline{I}}$ is the identity tensor. Total stress is related to strain $\underline{\underline{\varepsilon}}$ in an isotropically elastic material by the Navier-Poisson constitutive law

$$\underline{\underline{\sigma}} = -\frac{2\nu G}{1-2\nu}\,\mathrm{tr}(\underline{\underline{\varepsilon}}) - 2G\underline{\underline{\varepsilon}} \quad [6]$$

where tr( ) denotes the trace of a tensor, ν is the Poisson's ratio, and G is the shear modulus. Implicit in Eq. 6 is the definition

$$\langle \mathrm{tr}(\underline{\underline{\sigma}}) \rangle = p \quad [7]$$

where triangular brackets indicate an average over the number of spatial dimensions.

In the laboratory, G, which is the slope of the stress-strain curve at zero strain, can be measured by a torsion experiment. Poisson's ratio ν quantifies the ratio of lateral strain to axial strain and can obtained by a measurement of the bulk modulus K, which is the slope of a pressure *vs.* volume strain curve at zero volume strain. From G and K, the Poisson's ratio is computed by[12]

$$\nu = \frac{3K - 2G}{2(3K + G)} \quad [8]$$

In the regime of linear elasticity, deformations are considered sufficiently small that the strain can be written in terms of gradients of the displacement vector from an undeformed reference configuration, **u**, as

$$\underline{\underline{\varepsilon}} = \frac{1}{2}[\overline{\nabla}\mathbf{u} + (\overline{\nabla}\mathbf{u})^{\mathrm{T}}] \quad [9]$$

where a superscript T denotes the transpose of a tensor. Equations 4-6 and 9 summarize the equations governing an isotropic linear-elastic solid. Substitution of Eq. 8 into Eq. 6 and insertion of the result into Eq. 4 yields a system of differential equations in terms of the displacements and the Poisson's ratio

$$0 = \overline{\nabla} \cdot (\overline{\nabla}\mathbf{u}) + \frac{1}{1-2\nu}\overline{\nabla}(\overline{\nabla} \cdot \mathbf{u}) \quad [10]$$

Equations 4-10 are simplified forms of the equations governing linear elasticity; the general forms of these equations can be found in the text by Malvern.[13] We set out to solve Eq. 10 to obtain the system response to a periodic interfacial disturbance.

To develop boundary conditions on Eq. 10, we consider the case of a two-dimensional system (x- and z-directions) where the electrode surface is at $z = 0$ and the electrode and solvent extend infinitely far from this interfacial position. Then

$$u_x(x,\ z \to \pm\infty) = 0 \quad [11]$$

$$u_z(x,\ z \to \pm\infty) = 0 \quad [12]$$

At $z \geq 0$, the material is assumed to have properties characteristic of a solid polymer, G^s and ν^s; at all $z \leq 0$, it has properties of lithium, G^e and ν^e. The surface is subjected to a periodic displacement in the z-direction with amplitude A and frequency ω

$$u_z(x,\ 0) = A\cos(\omega x) \quad [13]$$

To obtain a physical solution in accordance with the surface force balance employed in the derivation of Eq. 3, tangential forces at the interface must balance[1] according to

$$\mathbf{e}_t(x,\ 0) \cdot \{\mathbf{e}_n(x,\ 0) \cdot [\underline{\underline{\tau}}_d^{\mathrm{electrode}}(x,\ 0) - \underline{\underline{\tau}}_d^{\mathrm{separator}}(x,\ 0)]\} = 0 \quad [14]$$

where $\mathbf{e}_t$ is a unit vector tangential to the surface.

To solve the problem posed by Eq. 10 and boundary conditions 11-13, displacement distributions satisfying Eq. 11-13 were computed in the separator ($z \geq 0$) and electrode ($z \leq 0$) regions. General solutions could be obtained by taking the functionality of the displacement distribution to be

$$\mathbf{u}(x,\ z) = \begin{bmatrix} u_x(x,\ z) \\ u_z(x,\ z) \end{bmatrix} = Re\left\{ e^{i\omega x}\begin{bmatrix} X(z) \\ Z(z) \end{bmatrix}\right\} \quad [15]$$

Insertion of form 15 into governing Eq. 10 separates varibles, leaving a system of two ordinary differential equations in z. Application of boundary conditions 11 and 12 for positive z gives a general solution valid in the separator region; their application for negative z gives a solution valid in the electrode. These two solutions can be matched by Eq. 13. After taking the real part of this result, the following set of displacement functions is obtained

$$\begin{aligned} u_x^s &= \left[A - \frac{(3-4\nu^s)}{\omega}B + Bz\right]\sin(\omega x)\exp(-\omega z) \\ u_z^s &= (A + Bz)\cos(\omega x)\exp(-\omega z) \\ u_x^e &= \left\{A - \frac{(3-4\nu^s)}{\omega}B + \left[\frac{2\omega}{(3-4\nu^e)}A - \frac{(3-4\nu^s)}{(3-4\nu^e)}B\right]z\right\}\sin(\omega x)\exp(\omega z) \\ u_z^e &= \left\{A - \left[\frac{2\omega}{(3-4\nu^e)}B - \frac{(3-4\nu^s)}{(3-4\nu^e)}B\right]z\right\}\cos(\omega x)\exp(\omega z) \end{aligned} \quad [16]$$

where B is an arbitrary constant, and superscripts s and e denote the separator and electrode regions, respectively. For further discussion of the procedure used to obtain Eq. 16, see the Appendix.

To determine a value of the constant B that makes Eq. 16 satisfy boundary condition 14, Eq. 5-7 and 9 can be employed to compute the deformation stresses on either side of the interface at $z = 0$. This procedure yields

$$\begin{aligned} \tau_{xx}^s(x,\ 0) &= 2G^s[-A\omega + 2(1-\nu^s)B]\cos(\omega x) \\ \tau_{xz}^s(x,\ 0) &= \tau_{zx}^s(x,\ 0) = 2G^s[A\omega - 2(1-\nu^s)B]\sin(\omega x) \\ \tau_{zz}^s(x,\ 0) &= 2G^s[A\omega - 2(1-\nu^s)B]\cos(\omega x) \end{aligned} \quad [17]$$

at the surface of the separator, and

$$\begin{aligned} \tau_{xx}^e(x,\ 0) &= \frac{2G^e}{(3-4\nu^e)} \times [-A\omega + 2(1-\nu^e)(3-4\nu^s)B]\cos(\omega x) \\ \tau_{xz}^e(x,\ 0) &= \tau_{xz}^e(x,\ 0) = \frac{2G^e}{(3-4\nu^e)} \times [-A\omega + 2(1-\nu^e)(3-4\nu^s)B]\sin(\omega x) \\ \tau_{zz}^e(x,\ 0) &= \frac{2G^e}{(3-4\nu^e)}[A\omega - 2(1-\nu^e)(3-4\nu^s)B]\cos(\omega x) \end{aligned} \quad [18]$$

at the surface of the electrode. Vectors normal and tangential to the surface were calculated from the displacement distributions given in Eq. 16 according to

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Table I. Parameters employed to evaluate current density at an interface between a lithium electrode and a polymer separator. Representative separator mechanical properties are given for solid crystalline PEO.[a,b]

$\gamma = 1.716$ J/m^2[14]	$G^s = 0.2$ GPa[16]	$\overline{V}_{LiX} = 1.674 \times 10^{-4}$ m^3/mol[18]
$G^e = 3.4$ GPa[15]	$\nu^s = 0.33$[17]	$t_-^0 = 0.7$[18]
$\nu^e = 0.42$[15]	$\overline{V}_{Li} = 1.3 \times 10^{-5}$ m^3/mol[14]	$\omega = 1 \times 10^8$ l/m[14]

[a] These PEO parameters overestimate by one or two orders of magnitude the shear moduli obtained for the amorphous polymer materials typically used as separators for lithium batteries.
[b] The shear modulus and Poisson's ratio of lithium were determined based on Young's modulus data from Ref. 15a and bulk modulus data from Ref. 15b. These were deconvoluted according to equations in Ref. 12.

$$\mathbf{e}_n = \begin{bmatrix} \dfrac{du_z(x,\ 0)}{dx} \\ 1 + \dfrac{du_x(x,\ 0)}{dx} \end{bmatrix} \Bigg/ \left\| \begin{bmatrix} \dfrac{du_z(x,\ 0)}{dx} \\ 1 + \dfrac{du_x(x,\ 0)}{dx} \end{bmatrix} \right\| \quad [19]$$

and

$$\mathbf{e}_t = \begin{bmatrix} 1 + \dfrac{du_x(x,\ 0)}{dx} \\ -\dfrac{du_z(x,\ 0)}{dx} \end{bmatrix} \Bigg/ \left\| \begin{bmatrix} 1 + \dfrac{du_x(x,\ 0)}{dx} \\ -\dfrac{du_z(x,\ 0)}{dx} \end{bmatrix} \right\| \quad [20]$$

where $\| \ \|$ denotes the magnitude of a vector. Equations 17-20 were substituted into boundary condition 14, which for small-amplitude displacements resulted in the relationship

$$B = A \frac{\omega[G^s(3 - 4\nu^e) + G^e]}{2[G^s(3 - 4\nu^e)(1 - \nu^s) + G^e(3 - 4\nu^s)(1 - \nu^e)]} \quad [21]$$

This expression is rigorously valid when $A = K_0/\omega$, where $K_0 \ll 1$. Substitution of Eq. 21 into Eq. 16 provides a solution to governing Eq. 10 for an interfacial disturbance of arbitrary amplitude and frequency.

Equations 16-21 and Eq. 5-7 and 9 were employed at $z = 0$ to give the relationships

$$\Delta p^{\text{electrode}} + \Delta p^{\text{separator}} = A\omega \left\{ \frac{G^{s^2}(3 - 4\nu^e) - G^{e^2}(3 - 4\nu^s)}{[G^s(3 - 4\nu^e)(1 - \nu^s) + G^e(3 - 4\nu^s)(1 - \nu^e)]} \right\} \cos(\omega x) \quad [22]$$

$$\mathbf{e}_n \cdot [\mathbf{e}_n \cdot (\underline{\underline{\tau}}^{\text{electrode}} - \underline{\underline{\tau}}^{\text{separator}})] = -A\omega \frac{4G^sG^e[1 - (\nu^s + \nu^e) + (1 - 2\nu^s)(1 - 2\nu^e)]}{[G^s(3 - 4\nu^e)(1 - \nu^s) + G^e(3 - 4\nu^s)(1 - \nu^e)]} \cos(\omega x) \quad [23]$$

which were necessary to evaluate the deformation parameter defined by Eq. 3. Also required was the product of surface energy and the surface gradient of the surface normal vector, which was determined for small-amplitude displacements to be

$$\gamma \overline{\nabla}_s \cdot \mathbf{e}_n = \gamma(\underline{\underline{I}} - \mathbf{e}_n\mathbf{e}_n) \cdot \overline{\nabla}\mathbf{e}_n = -A\omega^2\gamma \cos(\omega x) \quad [24]$$

Given the appropriate material property data, Eq. 2, 3, and 22-24 provide all information necessary to determine the current density at an interface deformed with a given frequency and small amplitude relative to that at an undeformed interface

$$\frac{i_{\text{deformed}}}{i_{\text{undeformed}}} = \exp\left[\frac{(1 - \alpha_a)\Delta\mu_{e^-}}{RT}\right] \quad [25]$$

Equation 25 demonstrates that if $\alpha_a < 1$ (typical for single-charge-transfer elementary redox reactions), current density at a deformed surface will be higher than that at an undeformed surface if $\Delta\mu_{e^-}$ is positive and lower if $\Delta\mu_{e^-}$ is negative. Because Eq. 22-24 have a similar dependence on x and A

$$\Delta\mu_{e^-} \propto A \cos(\omega x) \quad [26]$$

This argument leads to the conclusion that current density at a lithium "peak" ($A \cos(\omega x) > 0$) will be higher than that at a lithium "valley" ($A \cos(\omega x) < 0$) when

$$\frac{\Delta\mu_{e^-}}{A \cos(\omega x)} \equiv \Delta\mu > 0 \quad [27]$$

We refer to the quantity $\Delta\mu$ defined in Eq. 27 as the stability parameter. It should be evaluated in the limit as $A \to 0$, so that Eq. 21 rigorously satisfies the tangential-stress boundary condition. When the stability parameter is positive, electrode roughness tends to amplify (unstable deposition), because deposition on the peaks of deformed electrode surfaces is faster than on undeformed surfaces, and that in the valleys is slower. When the stability parameter is negative, electrode roughness tends to decay (stable deposition), because deposition on the peaks of deformed electrode surfaces is slower than on undeformed surfaces, and that in the valleys is faster. Examination of Eq. 25 also suggests that larger values of the stability parameter lead to higher exchange current densities at the peaks of roughened surfaces; a system with a high positive $\Delta\mu$ should therefore be expected to initiate dendritic deposits faster than one with a lower positive $\Delta\mu$.

Results and Discussion

Literature values of the material properties of a lithium/polymer system required for this analysis are shown in Table I.[a,14-18] The elastic properties listed for poly(ethylene oxide) (PEO) correspond to a high-molecular-weight, crystalline material. Practical polymer separators are typically composed of amorphous, shorter-chain polymers; while the Poisson's ratios of these materials are similar, the shear modulus may be lower by two or three orders of magnitude.[19]

Figure 1 shows the spatial position of material in the electrode and electrolyte phases yielded by combination of Eq. 16 and 21, given Poisson's ratios characteristic of crystalline PEO and lithium in the polymer and electrode regions, respectively, and an electrode shear modulus corresponding to that of lithium. These properties are

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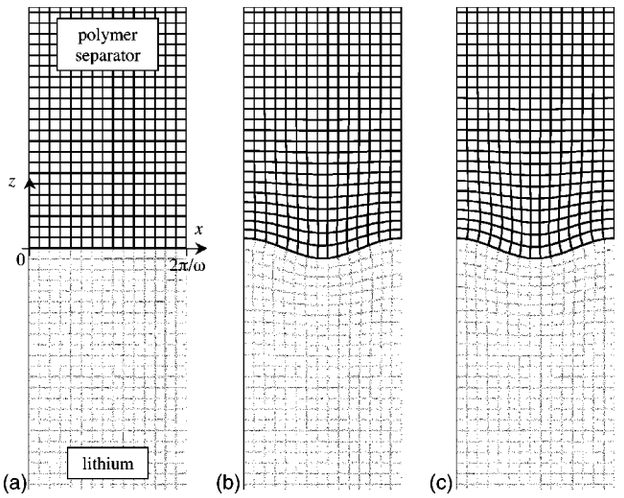


Figure 1. Displacement distributions yielded by Eq. 16 and 21 at different polymer shear moduli. One period of displacement in x and 1.5 displacement periods above and below the z axis are shown in each figure. The upper region has the material properties of lithium ($\nu^e = 0.42$, $G^e = 3.4$ GPa), and the lower has properties characteristic of a separator ($\nu^s = 0.33$, G^s differs). (a) The reference (undeformed) position $A = 0$, which is the same for all material properties in either region. (b) The displacement distribution for $A = 0.4/\omega$, $G^s = 10^{-9}\ G^e$. (c) $A = 0.4/\omega$, $G^s = 10\ G^e$.

listed in Table I. Horizontal gridlines in Fig. 1 correspond to the displaced position along lines of constant undeformed material position z; vertical gridlines give the displaced position along lines of constant undeformed material position x. Figure 1a shows the case where $A = 0$, the undeformed reference state. Figures 1b and c show the final locations of the material in Fig. 1a when the interface is periodically deformed with $A = 0.4/\omega$. This condition is close to the upper limit of validity of Eq. 21 but is chosen to make displacements visible in Fig. 1 and 2. From Fig. 1b to 1c, the polymer shear modulus is increased from $10^{-9}\ G^{Li}$, a polymer phase with minimal elastic stiffness, to a shear modulus ten times that of lithium. The shear modulus affects the slope of the vertical contour lines as they

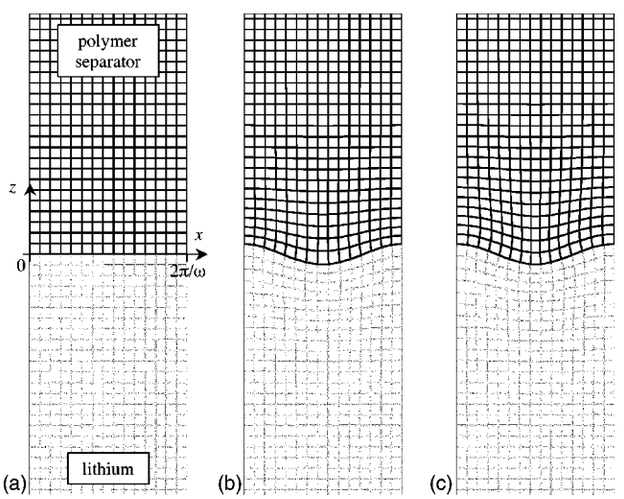


Figure 2. Displacement distributions yielded by Eq. 16 and 21 at differing Poisson's ratio. In each figure, $A = 0.4/\omega$ with $G = G^{Li}$ on both sides of the interface. (a) The undeformed material position; (b) a separator with $\nu^s = 0$, corresponding to a compressible material; and (c) a separator with $\nu^s = 0.5$, corresponding to an incompressible material.

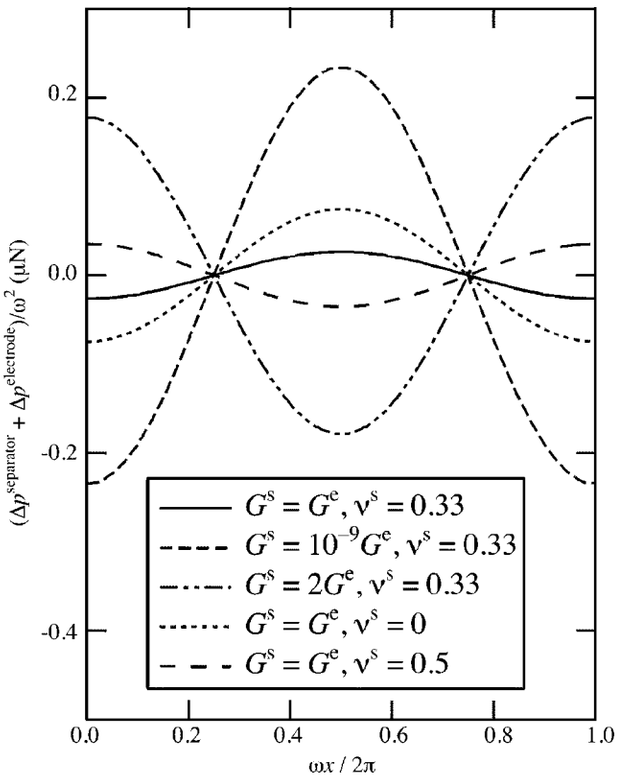


Figure 3. Gage pressure term from Eq. 22 with $A = 0.4/\omega$ for separators with various elastic properties. The black $G^s = G^e$, $\nu^s = 0.33$ curve is the case around which the others are varied.

hit the interface; the polymer elasticity in (b) leads to a narrower indentation of lithium for a given displacement amplitude and frequency than the higher shear modulus in (c).

Figure 2 demonstrates the effect of polymer Poisson's ratio on the periodic deformation profile at $A = 0.4/\omega$. Figure 2b shows the case of a fully compressible separator, $\nu^s = 0$, and Fig. 2c shows an incompressible separator, $\nu^s = 0.5$. The effect of compressibility can be seen most clearly by comparison of these figures to Fig. 2a. In the undeformed reference state, the horizontal and vertical grid lines mark out squares on the material. In Fig. 2b, the area between these gridlines is allowed to change in the separator material when the material deforms, because it is compressible. In Fig. 2c the area between gridlines in the deformed state remains identical to the area between gridlines in the undeformed state.

Although changes in the mechanical properties of the polymer material lead to small differences in the displacement distributions around the interface, they have a significant effect on the forces that act across it, as demonstrated by Fig. 3 and 4. Figure 3 shows the compressive forces acting across the periodically deforming interface, which are given as a function of x by Eq. 22. The displacement amplitude and frequency used to generate this figure are the same as those employed in Fig. 1 and 2. Both elastic properties of the separator can have a significant effect on compressive forces. In particular, it is significant that the sign of the compressive force amplitude can be reversed by changing either the separator Poisson's ratio (as in the $\nu^s = 0.5$ case), or the separator shear modulus (as when $G^s = 2G^e$). The deformation stresses acting across the interface, given by Eq. 23, are shown for the same property combinations in Fig. 4. In the deformation term, the amplitude has the same sign for all property combinations. Deformation stresses across the interface have a much larger amplitude when the separator material is fully compressible than when it is incompressible. When the separator

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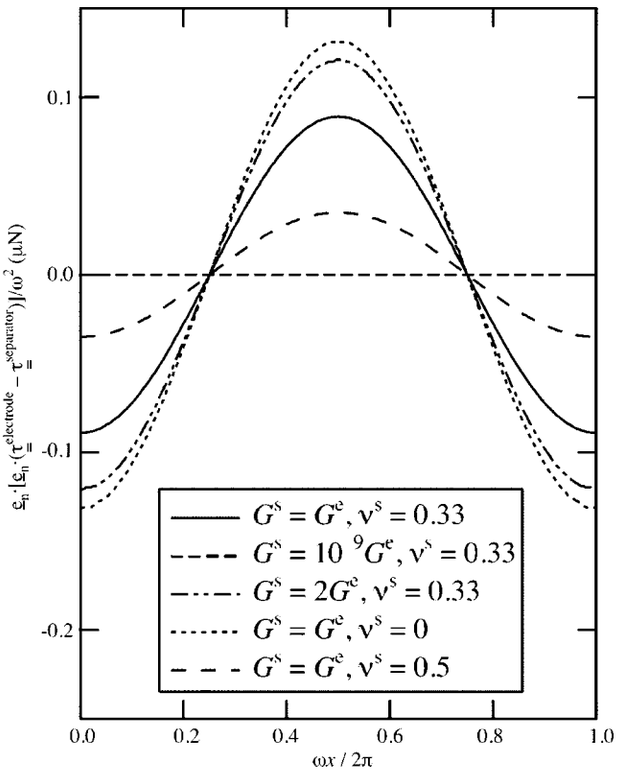


Figure 4. Deformation stress term from Eq. 23 with $A = 0.4/\omega$ for separators with various elastic properties. The black $G^s = G^e$, $\nu^s = 0.33$ curve is the case around which the others vary.

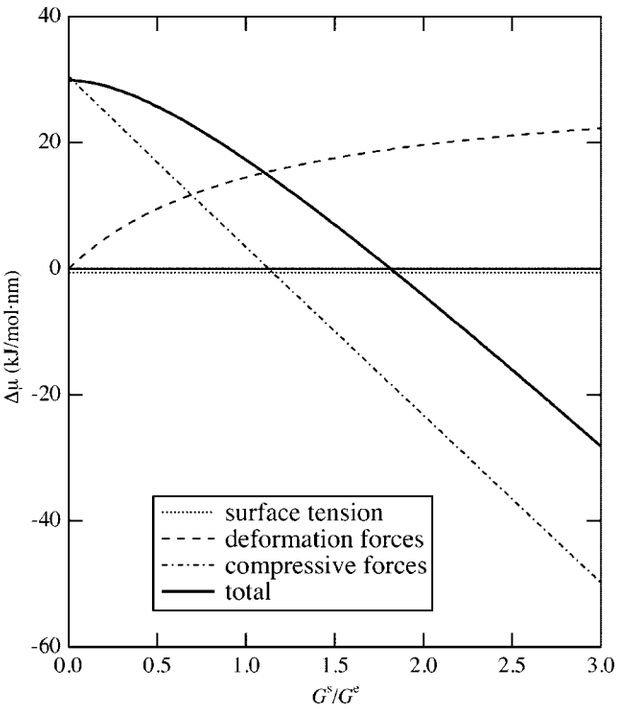


Figure 5. Contributions of compressive forces, deformational forces, and surface forces to the interfacial stability parameter as a function of separator shear modulus. Electrode properties are set to those of lithium, and the separator Poisson's ratio is that of PEO, $\nu^s = 0.33$.

shear modulus is near zero, the interface sustains almost no deformation stress; the amplitude of the deformation term increases with increasing separator shear modulus.

Surface tension is the final mechanical force that contributes to the chemical potential drop across the interface in this system. In the case where $A = 0.4/\omega$, the surface-force contribution is the only quantity that depends on the displacement frequency. Surface-force contributions are linearly proportional to the displacement frequency under these circumstances, while the compressive and deformational contributions are independent of it. These conditions imply that as the frequency of the displacement increases, *i.e.*, as the "valley-to-valley" distance across electrode protrusions lowers, the impact of surface forces increases relative to compressive or deformational effects. Some variability is therefore present in our model, as the frequency of interfacial roughening is unknown. However, microscopy experiments[20-22] and modeling[14] have shown that protrusions at roughening lithium electrodes are of the order 0.1-1 μm across. In this work, which focuses on mechanical effects besides the surface tension, a displacement frequency of $\omega = 2\pi/0.1\ \mu m \approx 10^8\ m^{-1}$ was assumed to be a reasonable choice of displacement frequency at the roughening cathode surface.

The contributions of compressive, deformational, and surface-tension forces to the stability parameter defined by Eq. 3 and 27 are shown as a function of separator shear modulus in Fig. 5. All data beside the polymer shear modulus used to generate this plot are listed in Table I. At shear moduli lower than about twice the shear modulus of the lithium electrode, the total stability parameter is positive, indicating electrode instability. Negative values of the total stability parameter, corresponding to stable electrodeposition, are obtained when the electrolyte shear modulus is greater than twice that of lithium. This transition to a stable growth regime is consistent with the inhibition of dendrites observed in systems where lithium electrodes are coated with conductive inorganic glasses; these materials typically have much higher shear moduli than polymer electrolytes.[23] In the figure, the total stability parameter is also broken into constituent contributions from surface, deformational, and compressive forces.

Surface tension promotes electrode stability, as shown by its negative contribution to the stability parameter. The surface-tension contribution is constant in Fig. 5 because it is independent of separator shear modulus. At the roughness frequency employed in this analysis, the magnitude of the surface-tension contribution is much smaller than those of the other mechanical contributions, demonstrating the greater relative impact of elastic solid properties on electrode stability when electrode roughening is on the micrometer scale.

Deformational forces contribute to electrode instability at all values of the separator shear modulus. Initially, the deformational-force contribution to instability rises rapidly with respect to G^s/G^e. Inspection of Fig. 5 and Eq. 23 shows that as the shear modulus of the separator rises, the deformational contribution to instability asymptotically approaches a limiting value. Because the properties of polymers presently employed in lithium battery systems are in the low G^s/G^e regime, this high-shear-modulus behavior is not shown on the figure. However, it is physically significant that though separator deformation always leads to system instability, its contribution levels off as G^s/G^e becomes larger.

The most noteworthy result visible from Fig. 5 is that compressive forces within the separator begin to contribute to stability as polymer shear modulus increases. As might be expected intuitively, compressive forces contribute to stability when the separator shear modulus exceeds that of lithium. In Fig. 5, the compressive force contribution to the stability parameter crosses zero at $G^s/G^e = 1.125$. This line does not cross zero at $G^s/G^e = 1$ because the Poisson's ratio of the separator is lower than that of the lithium metal.

To investigate the effect of separator compressibility further, the Poisson's ratio was varied in addition to the shear modulus. Figure 6

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gives the total stability parameter as a function of electrolyte shear modulus at three different values of the Poisson's ratio. The figure demonstrates that interfacial stability is obtained with an incompressible separator when $G^s/G^e = 1$. Stability is more difficult to achieve when the separator is compressible. In fact, the drawbacks of a compressible solid electrolyte outweigh its benefits when $\nu^s = 0$ and $G^s/G^e < 0.5$; the stability parameter actually rises with respect to separator shear modulus in this regime. Although the cause as the modulus of a viscoelastic polymer like PEO decreases, viscous forces within it become increasingly large relative to elastic ones. Regimes in which the neglect of viscous forces is justified can be identified by a straightforward dimensional analysis. Assuming that a peak on the roughening surface can be treated as a hemisphere growing from a flat surface and that surface growth is slow, the general formula for drag on a spherical object in an axisymmetric creeping flow[25] and Eq. 23 may be used to evaluate the forces due to viscosity and deformation acting on the peak. These formulas yield

$$\frac{f_{\text{viscous}}}{f_{\text{deformation}}} = O\left\{\frac{2\pi\mu^s v_{\text{peak}} r[G^s(3-4\nu^e)(1-\nu^s)+G^e(3-4\nu^s)(1-\nu^e)]}{4\pi r^2 G^s G^e[1-(\nu^s+\nu^e)+(1-2\nu^s)(1-2\nu^e)]}\right\} \quad [28]$$

Poisson's ratio is a difficult material property to control, an increase in system resistance to bulk compression can have a significant effect on electrode stability. Because hard particulate additives decrease the bulk compressibility of polymer materials, this result provides a mechanical explanation for why polymer electrolytes containing silica additives exhibit longer initiation times for dendritic deposits than additive-free polymer systems with similar shear moduli.[24]

Impact of Assumptions

The main assumption applied in this analysis is that the mechanical state of a roughening interface can be approximated by a 2D displacement profile. Future treatment of this stability problem should incorporate a 3D periodic displacement at the roughening interface. Because a 2D treatment of the mechanics has been employed here, the stability results displayed in Fig. 5 and 6 should be taken as accurate to only a first approximation.

Results of the elastic analysis have been extended to low values of the shear modulus with no consideration of additional internal mechanical forces. This extension of the model is questionable, be-

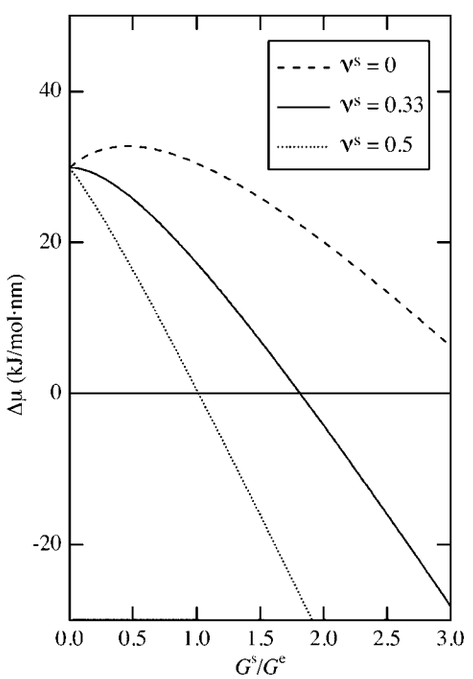


Figure 6. Interfacial stability parameter as a function of separator shear modulus and Poisson's ratio. Electrode properties are set to those of lithium.

where μ^s is the separator viscosity in the limit of zero shear rate, r is the radius of the hemispherical peak, v_{peak} is the peak velocity, and f is the force due to a particular kind of mechanical action. Assuming that G^s is small compared to G^e, that the Poisson's ratio of the separator is 0.4, and that the radius of the peak is of the same order as its wavelength, Eq. 28 reduces to

$$\frac{f_{\text{viscous}}}{f_{\text{deformation}}} = O\left\{\frac{\mu^s v_{\text{peak}}\omega}{G^s}\right\} \quad [29]$$

The analysis performed here rigorously applies under conditions when the ratio of viscous to deformational forces is small, or

$$G^s \gg \mu^s v_{\text{peak}}\omega \quad [30]$$

At currents approaching the limiting current, growth velocities tend to be on the order of 10 nm/s,[15] and the zero-shear-rate viscosities of most polymer electrolyte materials are on the order of 0.1 Pa s.[26] Equation 30 then shows that the analysis provided here, which neglects viscous forces, is strictly valid only when $G^s \gg 0.1$ Pa. This justifies the choice of the low-modulus limit $G^s = 10^{-9}\ G^e$.

An assumption of secondary importance was employed in the derivation of Eq. 21 from boundary condition Eq. 14. Terms of higher than first order in $A\omega$ and B were discarded on the basis that $A\omega$ and B were of similar magnitude and that both were much less than unity. This assumption is reasonable for the stability parameter, which is evaluated in the limit of infinitesimal-amplitude interfacial displacements; the most significant results of this work, presented in Fig. 5 and 6, represent cases where the linearization applies exactly. However, the condition that $A = 0.4/\omega$ that was employed to generate Fig. 1-4 is near the upper limit of validity of the linearization. It should be noted that if physical displacement distributions are to be obtained in the Hookean-elastic regime for larger values of A, the higher order nonlinearities that were discarded should be included when determining the relationship between B and A with the tangential-stress boundary condition. A more exact treatment valid at larger amplitudes could be obtained by numerical evaluation of higher order perturbations to the governing equations. It can be seen that the neglect of nonlinearities is significant at moderate amplitudes by setting the electrode and separator materials to have Poisson's ratios close to 0.5. In these cases, departure from the expected behavior, regions between gridlines in Fig. 2c maintaining similar areas to those in 2a, was observed when $A = 0.8/\omega$.

While it might appear restrictive, the assumption that the electrode and electrolyte are semi-infinite was determined to be reasonable. The exponential nature of the decay of displacements with respect to z in Eq. 16 shows that strain introduced by a displacement at the interface has decayed by 99% upon penetration depth of $z = \pm 4/\omega$ from the interface. Employing the value of ω shown in Table I, this corresponds to a distance of 40 nm. Because electrode

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spacings and electrode thicknesses in lithium/polymer systems are typically on the order of tens of micrometers, the semi-infinite treatment of electrode and electrolyte is appropriate.

The use of linear elasticity theory to develop displacement distributions could also introduce error. This theory is limited to very small strains, a restriction that is probably more significant than the neglect of higher order terms in boundary condition 14. Full treatment of system mechanics could extend the model to the nonlinear elasticity regime and include plastic deformation and stress relaxation. In addition, the fracture mechanics of the separator material have been ignored; separator yield or fracture could be important for brittle materials, which tend to have high shear moduli. Each of these issues would require a numerical treatment of the deformation problem. A finite-element analysis of the system with more sophisticated stress constitutive laws would be enlightening.

We should reiterate that local concentration variations have not been included in this model in order to provide as simple a perspective as possible on the role of mechanical forces in dendrite growth. The general kinetic model employed in this work does allow for the evaluation of such effects, which would be significant if transient modeling of roughening were performed. Because the analysis employed here involves surface displacements in a quasi-steady state, local concentration changes do not play a significant role. Concentration changes enter Eq. 2 through the $i_{0,\mathrm{ref}}$ term, and therefore all results generated here would scale similarly if the electrolyte concentration were changed.

One interesting physical property that has yet to be included in a theory of dendritic systems is the adhesive force that attaches the electrode to the electrolyte. In this model, it has been assumed that the electrode and electrolyte adhere perfectly. However, in systems at low pressure, separator delamination has been observed as dendrites begin to grow.[27] The quality of the electrode/electrolyte contact can be easily evaluated by measurements of low-frequency impedance; in cells where contact is poor, low-frequency impedance tends to be highly variable. Evidence of delamination in cells at ambient pressure is not abundant.

Conclusion

Each of the assumptions employed here limits the accuracy of this analysis to some extent, and the presented results should therefore be interpreted from a qualitative perspective. Linear elasticity theory was employed to generate physical pictures of periodically deformed interfaces. The displacement distributions generated by this analysis were used to evaluate the impact of polymer shear modulus and Poisson's ratio on the shapes of periodically roughened lithium/polymer interfaces. Although changes in elastic parameters of the separator led to small differences in the resulting displacement distributions, they had a great effect on the forces at the interface.

Surface tension was shown in roughening systems with solid electrolytes to contribute minimally to electrode stability in comparison to the much larger impacts of compressive and deformational forces acting across the interface. However, surface tension forces always contributed to stability. The deformational forces across the interface had a destabilizing effect that increased with the shear modulus of the separator material. Deformational contributions to the stability parameter were found to approach a maximum value asymptotically as separator shear modulus increased. Compressive forces contributed increasingly to stability as shear modulus increased. When the net impact of all three mechanical forces was evaluated, it was found that a transition to electrode stability occurred at separator shear moduli of the same order of magnitude as the electrode modulus.

The effect of electrolyte compressibility on stability was also evaluated. It was shown that systems with a greater degree of polymer compressibility were less stable. Stability was achieved at the lowest polymer shear modulus with an incompressible separator. Decreasing the Poisson's ratio to zero increased the transition to stability by a factor of four in shear modulus.

A theoretical justification for the experimental observation of dendritic suppression in mechanically enhanced systems was provided by this analysis. Separator mechanical properties were allowed to vary, showing that conditions exist under which the instability of lithium/polymer interfaces can be eliminated by mechanical means. This regime of stability occurs at separator shear moduli three or more orders of magnitude higher than those achieved by commonly researched polymer electrolytes.

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Appendix

Here we present the procedure used to calculate Eq. 16, which are solutions to Eq. 10 that satisfy boundary conditions 11, 12, and 13. In two dimensions, Eq. 10 becomes

$$x\text{ - direction:}0 = \frac{\partial^2 u_x}{\partial x^2} + \frac{\partial^2 u_x}{\partial z^2} + \frac{1}{(1-2\nu)}\left[\frac{\partial^2 u_x}{\partial x^2} + \frac{\partial^2 u_z}{\partial x \partial z}\right] \quad [A\text{-}1]$$

$$z\text{ - direction:}0 = \frac{\partial^2 u_z}{\partial x^2} + \frac{\partial^2 u_z}{\partial z^2} + \frac{1}{(1-2\nu)}\left[\frac{\partial^2 u_x}{\partial x \partial z} + \frac{\partial^2 u_z}{\partial z^2}\right] \quad [A\text{-}2]$$

Insertion of the complex argument of $Re\{\,\}$ from Eq. 15 into Eq. A-1 and A-2 gives

$$x\text{ - direction:}0 = -\omega^2 X + \frac{d^2 X}{dz^2} + \frac{1}{(1-2\nu)}\left[-\omega^2 X + i\omega \frac{dZ}{dz}\right] \quad [A\text{-}3]$$

$$z\text{ - direction:}0 = -\omega^2 Z + \frac{d^2 Z}{dz^2} + \frac{1}{(1-2\nu)}\left[i\omega \frac{dX}{dz} + \frac{d^2 Z}{dz^2}\right] \quad [A\text{-}4]$$

Equation A-3 and A-4 are second-order ordinary differential equations in z, and are amenable to Laplace transform. Taking the Laplace transform with respect to z shows that

$$0 = 2(1-\nu)\omega^2 \bar{X} - (1-2\nu)[s^2 \bar{X} - sX(0) - X'(0)] - i\omega[s\bar{Z} - Z(0)] \quad [A\text{-}5]$$

$$0 = \omega^2(1-2\nu)\bar{Z} - 2(1-\nu)[s^2 \bar{Z} - sZ(0) - Z'(0)] - i\omega[s\bar{X} - X(0)] \quad [A\text{-}6]$$

where the barred variables indicate transformed functions and s is the transformed z-coordinate. Primes indicate differentiation with respect to z. Equation A-5 and A-6 can be written more concisely in matrix form as

$$\begin{bmatrix} (1-2\nu)s^2 - 2(1-\nu)\omega^2 & i\omega s \\ i\omega s & 2(1-\nu)s^2 - (1-2\nu)\omega^2 \end{bmatrix}\begin{bmatrix} \bar{X} \\ \bar{Z} \end{bmatrix} = \begin{bmatrix} (1-2\nu)[sX(0) + X'(0)] + i\omega Z(0) \\ 2(1-\nu)[sZ(0) + Z'(0)] + i\omega X(0) \end{bmatrix} \quad [A\text{-}7]$$

Evaluating the determinant of the square matrix

$$\mathbf{A} = \begin{bmatrix} (1-2\nu)s^2 - 2(1-\nu)\omega^2 & i\omega s \\ i\omega s & 2(1-\nu)s^2 - (1-2\nu)\omega^2 \end{bmatrix} \quad [A\text{-}8]$$

we find

$$\det(\mathbf{A}) = 2(1-\nu)(1-2\nu)(s^2-\omega^2)^2 \quad [A\text{-}9]$$

which provides insight toward the form of the solution.

If Cramer's rule is applied to Eq. A-7, the numerator of each resulting root contains no term of order higher than s^3, and each denominator is the determinant shown in Eq. A-9 (order s^4). Therefore the transform is well-behaved, and when $\det(\mathbf{A}) = 0$ there are simple poles in Laplace space. The poles are of at most second order at $s = \pm\omega$, so it is expected from the theorem of residues that the inverted solution has the form

$$\begin{bmatrix} X(z) \\ Z(z) \end{bmatrix} = \begin{bmatrix} C_1^X e^{-\omega z} + C_2^X z e^{-\omega z} + C_3^X e^{\omega z} + C_4^X z e^{\omega z} \\ C_1^Z e^{-\omega z} + C_2^Z z e^{-\omega z} + C_3^Z e^{\omega z} + C_4^Z z e^{\omega z} \end{bmatrix} \quad [A\text{-}10]$$

This topological analysis can be used to find a general solution for any system of n ordinary differential equations where the Laplace transform is polynomial in s. Note

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that the differential Eq. A-3 and A-4 are second-order, so only four undetermined constants should be expected in a complete solution; linear independence of each of the terms in Eq. A-10 suggests that the constraint $C_i^X = f(C_i^Z)$ can be applied to satisfy this requirement.

We illustrate the procedure by which the number of constants is reduced by treating the region where z is positive. This region obeys boundary conditions 11 and 12 in the positive z-direction

$$u_x(x,\ z \to \infty) = 0 \qquad \text{[A-11]}$$

$$u_z(x,\ z \to \infty) = 0 \qquad \text{[A-12]}$$

In this case, the last two constants in both vector components of Eq. A-10, C_3^i, and C_4^i can immediately be discarded because the terms containing them diverge as z tends to infinity. The real part of the fundamental solution becomes

$$\begin{bmatrix} u_x(x,\ z) \\ u_z(x,\ z) \end{bmatrix} = Re\left\{ e^{i\omega x} \begin{bmatrix} C_1^X \exp(-\omega z) + C_2^X z \exp(-\omega z) \\ C_1^Z \exp(-\omega z) + C_2^Z z \exp(-\omega z) \end{bmatrix} \right\} \qquad \text{[A-13]}$$

which can be used to find additional relationships between the undetermined constants.

C_i^Z must be real to make the z-displacement have pure cosine functionality in x, but C_i^X can contain imaginary components. After writing C_i^X in a form that reveals their complex character

$$\begin{aligned} C_1^X &= a_1^X - ib_1^X \\ C_2^X &= a_2^X - ib_2^X \end{aligned} \qquad \text{[A-14]}$$

insertion into Eq. A-13 shows that

$$\begin{bmatrix} u_x(x,\ z) \\ u_z(x,\ z) \end{bmatrix} = Re\left\{ e^{i\omega x} \begin{bmatrix} (a_1^X - ib_1^X)\exp(-\omega z) + (a_2^X - ib_2^X) z \exp(-\omega z) \\ C_1^Z \exp(-\omega z) + C_2^Z z \exp(-\omega z) \end{bmatrix} \right\} \qquad \text{[A-15]}$$

Taking the real part of Eq. A-15 gives

$$\begin{bmatrix} u_x(x,\ z) \\ u_z(x,\ z) \end{bmatrix} = \cos(\omega x)\exp(-\omega z) \begin{bmatrix} a_1^X + a_2^X z \\ C_1^Z + C_2^Z z \end{bmatrix} + \sin(\omega x)\exp(-\omega z) \begin{bmatrix} b_1^X + b_2^X z \\ 0 \end{bmatrix} \qquad \text{[A-16]}$$

If Eq. A-16 is substituted into Eq. A-1 and A-2, the results are

$$\begin{aligned} 0 = {} & [a_1^X \omega + 2(1-2\nu)a_2^X + a_2^X \omega z]\cos(\omega x) \\ & + \{(C_1^Z - b_1^X)\omega - [2(1-2\nu)b_2^X + C_2^Z] + (C_2^Z - b_2^X)\omega z\}\sin(\omega x) \end{aligned} \qquad \text{[A-17]}$$

$$\begin{aligned} 0 = {} & \{(C_1^Z - b_1^X)\omega + [b_2^X - 4(1-\nu)C_2^Z] \\ & + (C_2^Z - b_2^X)\omega z\}\cos(\omega x) + (a_1^X \omega - a_2^X + a_2^X \omega z)\sin(\omega x) \end{aligned} \qquad \text{[A-18]}$$

The cosine and sine functions are linearly independent, so Eq. A-17 and A-18 result in a system of four equalities that must be satisfied by the constants

$$\begin{aligned} 0 &= a_1^X \omega + 2(1-2\nu)a_2^X + a_2^X \omega z \\ 0 &= (C_1^Z - b_1^X)\omega - [2(1-2\nu)b_2^X + C_2^Z] + (C_2^Z - b_2^X)\omega z \\ 0 &= (C_1^Z - b_1^X)\omega + [b_2^X - 4(1-\nu)C_2^Z] + (C_2^Z - b_2^X)\omega z \\ 0 &= a_1^X \omega - a_2^X + a_2^X \omega z \end{aligned} \qquad \text{[A-19]}$$

Constant terms are linearly independent of those multiplied by z, allowing further separation of Eq. A-19. This results in a system of six equations in our six unknown coefficients; six equations, not eight, result, because two are redundant

$$\begin{aligned} 0 &= a_1^X \omega + 2(1-2\nu)a_2^X \\ 0 &= a_2^X \omega \\ 0 &= a_1^X \omega - a_2^X \\ 0 &= C_2^Z - b_2^X \\ 0 &= (C_1^Z - b_1^X)\omega - [2(1-2\nu)b_2^X + C_2^Z] \\ 0 &= (C_1^Z - b_1^X)\omega + [b_2^X - 4(1-\nu)C_2^Z] \end{aligned} \qquad \text{[A-20]}$$

From Eq. A-20, the following relationships among constants are deduced

$$\begin{aligned} a_1^X &= 0 \\ a_2^X &= 0 \\ b_2^X &= C_2^Z \\ b_1^X &= C_1^Z - \frac{(3-4\nu)}{\omega} C_2^Z \end{aligned} \qquad \text{[A-21]}$$

We have thus achieved a result for the periodic basis solution for positive z

$$\mathbf{u}(x,z;z \geqslant 0) = \begin{bmatrix} u_x \\ u_z \end{bmatrix} = \begin{bmatrix} \left(C_1^Z - \frac{3-4\nu}{\omega} C_2^Z + C_2^Z z \right)\sin(\omega x) \\ (C_1^Z + C_2^Z z)\cos(\omega x) \end{bmatrix} \exp(-\omega z) \qquad \text{[A-22]}$$

A similar procedure to that illustrated can be used to obtain a general basis solution for negative z. This solution satisfies boundary conditions 11 and 12 as z tends to negative infinity and therefore has null values of C_1^i and C_2^i. The cosine functionality of the boundary condition mandates that C_3^Z is real. The procedure used from Eq. A-14 through A-22 can then be used under the assumption that C_i^X has complex character. The resulting basis solution for negative z has the form

$$\mathbf{u}(x,\ z;\ z \leqslant 0) = \begin{bmatrix} u_x \\ u_z \end{bmatrix} = \begin{bmatrix} -\left(C_3^Z + \frac{(3-4\nu)}{\omega} C_4^Z + C_4^Z z \right)\sin(\omega x) \\ (C_3^Z + C_4^Z z)\cos(\omega x) \end{bmatrix} \exp(\omega z) \qquad \text{[A-23]}$$

Equation A-22 and A-23 can then be matched at $z = 0$, according to the restrictions that

$$C_3^Z = C_1^Z = A \qquad \text{[A-24]}$$

and that

$$u_x(x,0;z \leqslant 0) = u_x(x,0;z \geqslant 0) \qquad \text{[A-25]}$$

bearing in mind that the Poisson's ratios in different regimes of z differ. The result is Eq. 16.

List of Symbols

a_1	real constant defined in Eq. A-14, m
a_2	real constant defined in Eq. A-14, l
$\mathbf{A}$	square matrix defined in Eq. A-8, l/m^2
A	amplitude of interfacial perturbation, m
b_1	real constant defined in Eq. A-14, m
b_2	real constant defined in Eq. A-14, l
B	arbitrary constant, l
C_1, C_3	complex constants defined in Eq. A-10, m
C_2, C_4	complex constants defined in Eq. A-10, l
$\mathbf{e}$	surface unit vector, l
f	force, N
F	Faraday's constant, 96,487 C/mol
G	shear modulus, N/m^2
i_n	current density normal to the interface, A/m^2
$i_{0,\mathrm{ref}}$	exchange current density, A/m^2
$\underline{\underline{I}}$	identity tensor, l
K	bulk modulus, N/m^2
p	gage pressure, N/m^2
r	radius, m
R	ideal gas constant, 8.3143 J/mol K
s	Laplace-transformed independent variable, l/m
t_-^0	anion transference number, l
T	temperature, K
$\mathbf{u}$	displacement vector, m
$\underline{v}$	velocity, m/s
$\overline{V}_i$	partial molar volume, m^3/mol
x	x-coördinate, m
X	function used to separate variables in Eq. 10, defined in Eq. 15, m
z	z-coördinate, m
Z	function used to separate variables in Eq. 10, defined in Eq. 15, m

Greek

α	apparent transfer coefficient, l
γ	surface energy, J/m^2
$\underline{\underline{\varepsilon}}$	strain tensor, l
η_s	surface overpotential, V
$\Delta\mu_{e^-}$	deformation factor representing chemical potential change of electrons in the lithium electrode due to interfacial deformation, J/mol
$\Delta\mu$	stability parameter defined in Eq. 27, J/mol m

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μ viscosity in the limit of zero shear rate, kg/m s
ν Poisson's ratio, l
$\underline{\underline{\sigma}}$ total stress tensor, or Cauchy stress tensor, N/m^2
$\underline{\underline{\tau}}_d$ deformation stress tensor, N/m^2
ω frequency of interfacial perturbation, l/m

Subscripts and superscripts

a anodic
c cathodic
d deformation
e electrode
Li lithium
LiX lithium salt
n normal
s (subscript) surface
s (superscript) separator region
t tangential
x *x*-direction
X corresponding to *X*
z *z*-direction
Z corresponding to *Z*

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